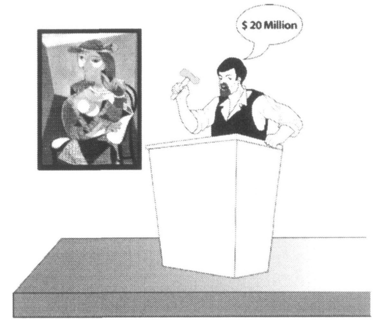
Chapter 1.

Time series data.

Situation One.



In the year 2015 Mrs Maggie Smithson sells a valuable painting at auction for $1 243 000.

Mrs Smithson's accountant informs her that for taxation purposes he needs to know what the value of the painting was when it first came into her possession.

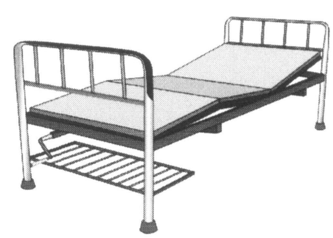
Mrs Smithson says that it was left to her in 2001 upon the death of the previous owner, Mrs Smithson's aunt, but she did not know the value of the painting at that time. What she did have though was a number of valuations, carried out at various times for insurance purposes.

The accountant listed these insurance valuations, together with the price achieved at auction as follows:

Year of valuation	1948	1968	1988	2008	2015
Valuation	$85 000	$280 000	$345 000	$810 000	$1 243 000

Suggest what you consider to be a reasonable valuation for the painting for the year that it first became the property of Mrs Smithson and explain how you arrived at your estimate and why you think it reasonable.

Situation Two.



A group is asked to write a report outlining the future health needs for a particular region. One of the things the group needs is an estimate of the number of hospital beds the region is likely to need in 5, 10 and 15 years time.
The number of hospital beds available in the region over the last twenty years, recorded every 2 years, is known and is shown in the following table:

Time	20 yrs ago	18 yrs ago	16 yrs ago	14 yrs ago	12 yrs ago	10 yrs ago
Number of hospital beds	3270	3350	3510	3520	3760	3900

Time	8 yrs ago	6 yrs ago	4 yrs ago	2 yrs ago	Now
Number of hospital beds	4100	4420	4790	5140	5600

Assuming the above numbers did just about meet the needs of the community at the time use the figures to make predictions for the number of hospital beds this region is likely to require in 5, 10 and 15 year's time and explain how you arrived at your predictions.

The situations of the previous page each involve measurements of something where the time the measurement is made is also recorded. In each situation the given information is an example of **time series** data.

Each of the situations involved the statistical investigation process:

① Clarify the problem and formulate appropriate questions.

In Situation One we needed to determine the value of the painting when it came into Mrs Smithson's possession. Not knowing this information we needed to ask what was known about its value at other times.

In Situation Two we needed to determine the number of hospital beds required for this region in 5, 10 and 15 year's time. To determine this we needed to ask for historical data about the number of beds required in the region.

② Plan the collection and use of appropriate data.

In Situation One the plan was to use previous insurance valuations we were given.

In Situation Two the plan was to use previous data that was available regarding bed numbers.

③ Apply appropriate techniques to analyse the data.

This was your task.

Did you consider graphing the information given in each of the situations?

Graphing time series data can allow underlying **trends** to be seen and predictions to be made. (The trend is the general direction of the time series, eg – increasing, decreasing, etc.)

④ Interpret the results in order to answer the original question and communicate your findings.

This also was your task.

If you did graph the data of the two situations you may have come to the conclusion that attempting to fit a straight line to the data might not be a suitable choice of "model". Perhaps you tried another model or, with each set of points perhaps suggesting a suitable curve that could be drawn "freehand" maybe you drew such a curve and read off the required values, or maybe you simply joined adjacent points with straight lines.

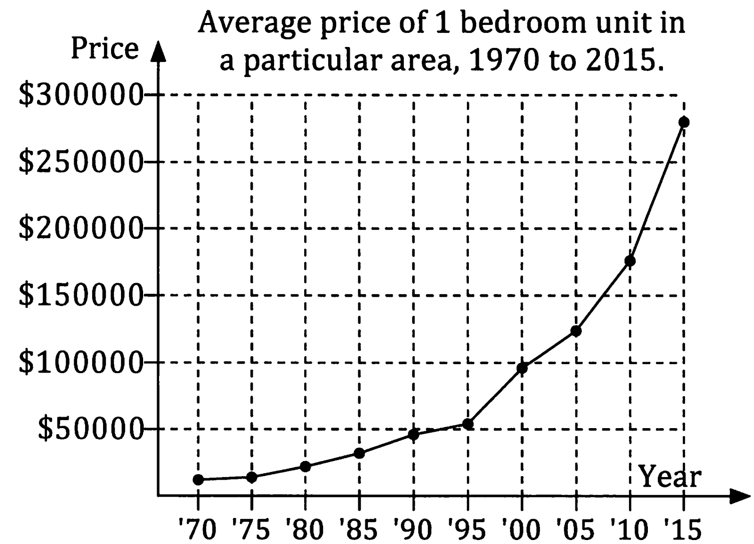
One of the situations required us to **extrapolate** beyond the known values so predictions obtained in this way would have to be viewed with caution. However, though extrapolation can be unreliable, it is sometimes the best we can do. With the situation that did not require extrapolation, but instead required us to determine a value between known points (**interpolation**), our prediction should be more reliable. However, even with interpolation, out of the ordinary events and short term fluctuations can sometimes give values that are not in line with normal trend. For example we might expect monthly visitor numbers to a city to peak higher than a general trend would suggest if the city were to host an Olympic Games for a month. We might expect daily traffic flow along a particular major road to drop below the figure a general trend would suggest if there were major roadworks taking place along the road and diversions were in place. Etc.

Time series data.

The diagram on the right shows a **line graph**. I.e a graph in which each point is joined to the next by a straight line.

This particular line graph involves observations that are ordered according to time, in this case in five year intervals from 1970 to 2015. The graph involves **time series data**.

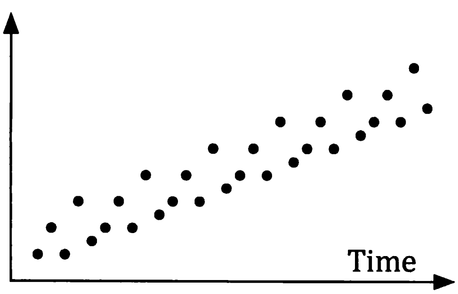
Note that when graphing time series data, time is plotted on the horizontal axis.

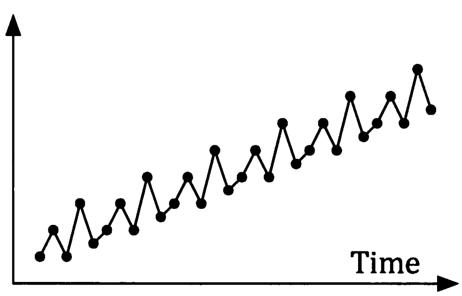


The joining of the points allows the underlying trend to be seen.

The graph shows an **increasing trend** – the prices increase with the passing of time.

Drawing straight lines between adjacent points can allow underlying patterns to become more evident. In the graph shown below there is clearly an increasing trend but can you also see the five point repeating pattern?





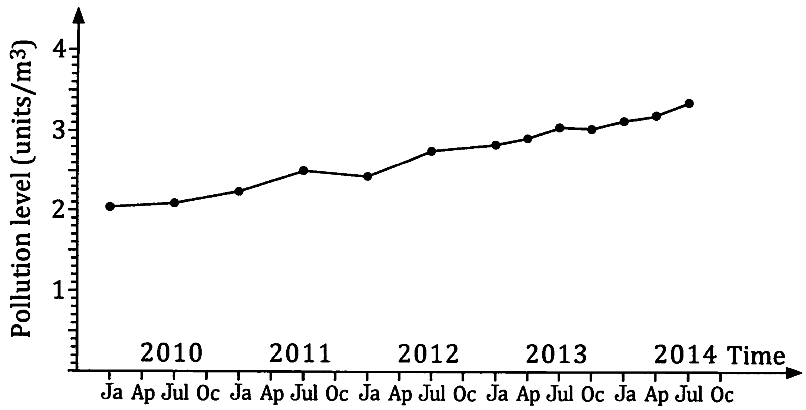
We will consider repeating patterns like this in more detail in the next chapter.

It can sometimes be the case that time series data is collected at intervals that are not all of the same length (as was the case in Situation One). Consider the following situation for example:

An environmental group, concerned with the levels of a particular pollutant in a river, collected and analysed samples from the river over a period of some years. The results are shown in the table below.

Month and Year	Jan 2010	Jul 2010	Jan 2011	Jul 2011	Jan 2012	Jul 2012	Jan 2013	Apr 2013	Jul 2013	Oct 2013	Jan 2014	Apr 2014	Jul 2014
Pollutant level (units / m^3)	2·04	2·09	2·24	2·51	2·44	2·76	2·84	2·92	3·06	3·04	3·14	3·21	3·37

Notice that in 2010, 2011 and 2012 levels were recorder half yearly, but in 2013 levels were recorded quarterly. The graph of this data is shown on the next page.



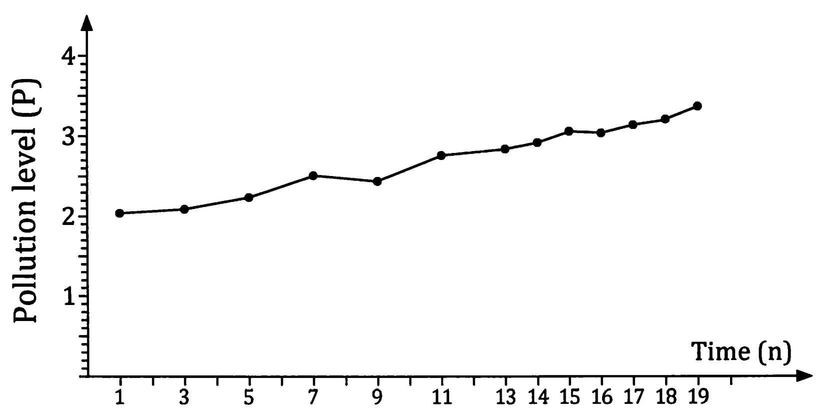
☞ Notice that the horizontal axis uses a consistent time scale throughout. The fact that from January 2013 onwards pollutant levels are given 4 times per year, rather than the two per year prior to this, changes the number of points plotted per year not the scale on the axis.

Making Predictions.

If we want to analyse the data mathematically we might need to number the dates involved, rather than to attempt putting Jan 2010, Jul 2010 etc into our calculator.
Particular care needs to be taken if the frequency of data collection changes as in this pollution levels example. Noticing that from 2013 onwards pollutant levels are given 4 times per year we could number the data points as follows:

Data Point (n)	1	3	5	7	9	11	13	14	15	16	17	18	19
Month and Year	Jan 2010	Jul 2010	Jan 2011	Jul 2011	Jan 2012	Jul 2012	Jan 2013	Apr 2013	Jul 2013	Oct 2013	Jan 2014	Apr 2014	Jul 2014
Pollution (P) (units / m^3)	2·04	2·09	2·24	2·51	2·44	2·76	2·84	2·92	3·06	3·04	3·14	3·21	3·37

The graph would then be as shown below:



With the data following a reasonably straight line we could then use our knowledge of linear regression to determine the least squares regression line:

$$P = 0{\cdot}0735n + 1{\cdot}91$$

(and a correlation coefficient of 0·99.)

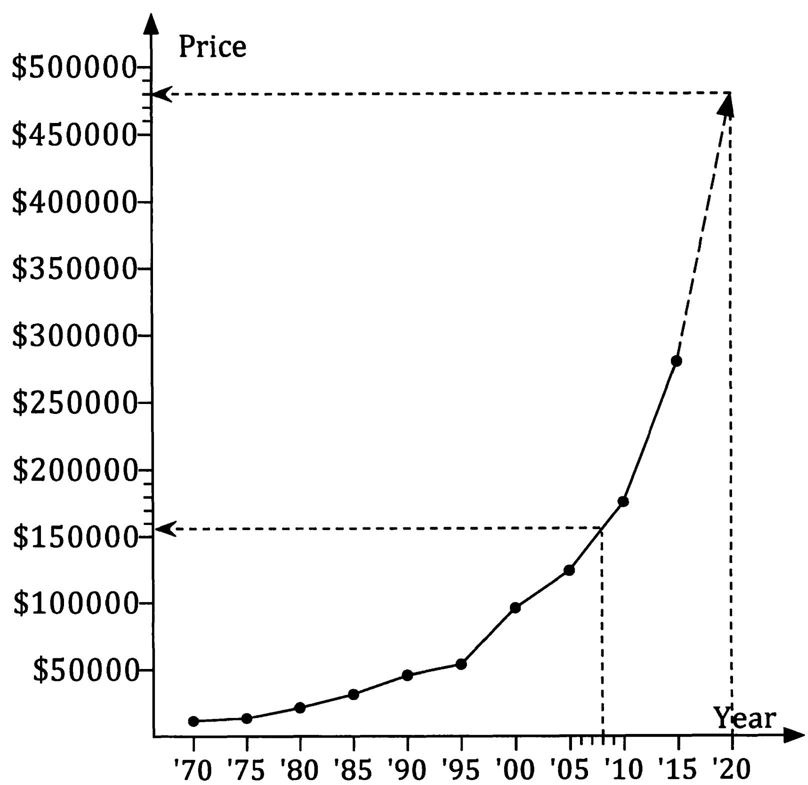
Asked to predict pollution levels for January 2018 we note that this would be for when n has the value 33.

For this value of n the regression line gives a predicted value of 4·3 for P.

Thus the predicted pollution level for January 2018 is 4·3 units / m^3.
However, even with our "eye-balling" of the points as being reasonably linear, and the correlation coefficient of 0·99 indicating that using a linear model is appropriate, January 2018 is well beyond dates during which pollution readings were collected so the reliability of the prediction is extremely questionable. If the trend continues it may be a reasonable prediction but from July 2014 to Jan 2018 many things could occur to alter the trend.

Of course in some cases linear regression may not be suitable – as in the two situations at the beginning of this chapter. In such cases other models available on some calculators could be used, and you are encouraged to explore such possibilities, or simply "eye-ball" the data and continue the trend as seems appropriate. Whilst in this text we will concentrate on using a least squares linear regression model it is clearly important to be able to recognise situations for which linear regression is not suitable.

Consider again the earlier situation involving the price of a 1 bedroom unit in a particular area. In this case linear regression would be inappropriate because a straight line model would not fit the data points at all well.
If asked to use the graph to predict what the average price was likely to have been in 2008 we could draw the line up from 2008 until it meets our graph and then draw across, as shown on the right, to obtain an estimate, in this case approximately \$156 000.



Similarly, if asked to estimate the likely average 1 bed unit price for the year 2020 we could continue the trend of the graph, as best we can, and read off a value, in this case approximately \$480 000.
However, as we know, this latter situation again involves extrapolation so any prediction could not be considered to be particularly reliable.

Example 1

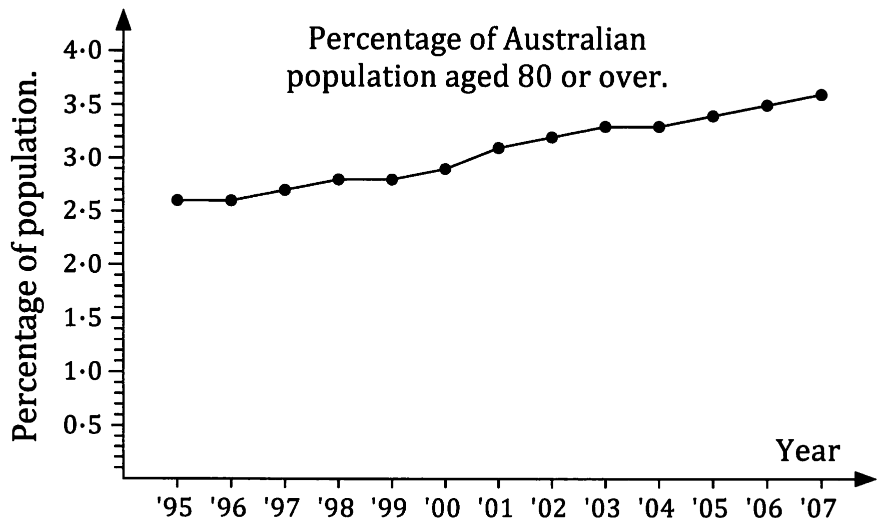
The table below shows the percentage of the Australian population aged 80 or over, for the years 1995 to 2007.

Percentage of the Australian population aged 80 and over.													
Year	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007
Percentage	2·6	2·6	2·7	2·8	2·8	2·9	3·1	3·2	3·3	3·3	3·4	3·5	3·6

[Source of data: Australian Bureau of Statistics.]

Display this information as a line graph and comment on any trends shown.
Use your graph to suggest what percentage of the Australian population will be aged 80 or over in the year 2020 and in the year 2050, each time discussing the likelihood of your prediction being reliable.

The graph is shown below.



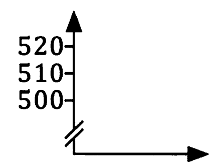
The percentage of the Australian population aged 80 or over shows an increasing trend. For the period the data is for, i.e. 1995 to 2007, the percentage increases fairly steadily at a rate of approximately 0·1 % of the Australian population per annum. (Figures show increase of 1% of population in 12 years.)
Continuing the trend forward to 2020, just over another 12 years, would suggest approximately 4·6% or 4·7% of the population would be 80 or over by then. With the data showing a reasonably linear trend we could use linear regression to obtain a 2020 figure of approximately 4·7%.
This prediction involves extrapolation so the result should not be considered particularly reliable but considering the steadiness of the increase from 1995 to 2007 the estimate may prove to be quite good.
The reasonably steady increase of about 1% every 12 years shown in the graph suggest approximately 7·2% of the population would be 80 or over by 2050. (Linear regression gives 7·4%.) However this requires us to extrapolate so far beyond the known points the "prediction" is perhaps more an "educated guess" than a reliable prediction.

→ Try to obtain more up to date percentages and see how they compare. ←

Note

- Time series data can often be *seasonal* in nature. For example, in the pollution levels in a river situation encountered earlier, the higher temperatures in summer may alter the levels of some pollutants. Thus in addition to showing some overall trend this sort of data may also show variation dependent upon the *season* when the data is collected. We will consider the seasonal nature of some time series data in the next chapter.

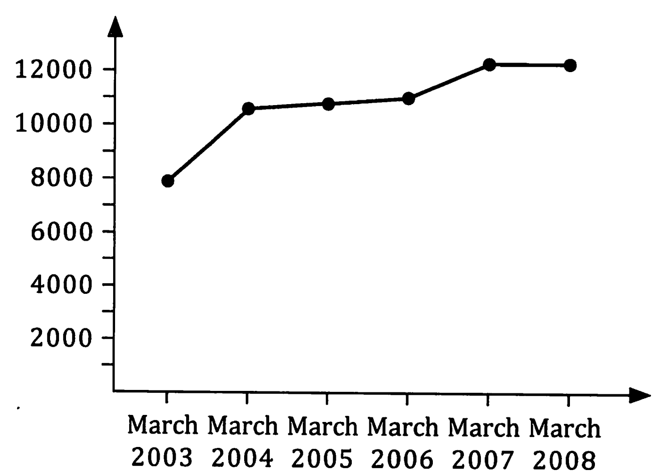
- The vertical axis should start from zero if possible. Sometimes this may be inconvenient or impractical and in such cases a break in the axis should be clearly shown (see the diagram on the right).



- Remember that time is plotted on the horizontal axis and that you should use the same time scale along the entire axis.

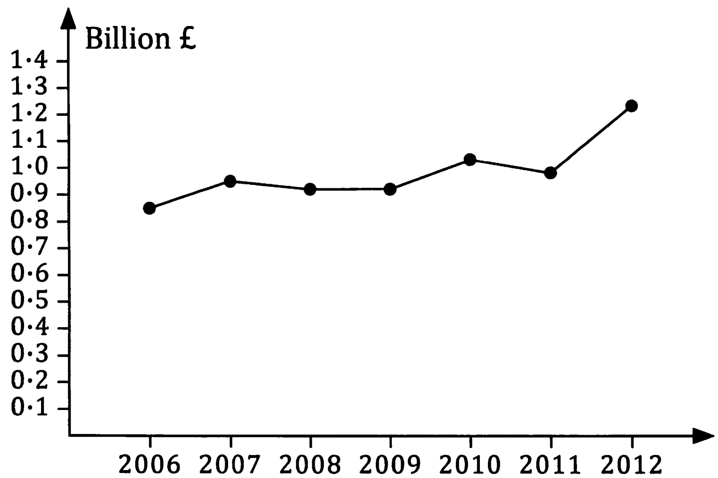
- If you have access to a computer with spreadsheet capability explore its ability, and the ability of your calculator, to draw line graphs and use them to draw some of the graphs requested in the next exercise.

- Remember that predicting into the future can be hazardous. Unanticipated events can occur. Consider for example the graph below, which shows the Dow Jones Index (a measure of the share values of a number of large companies in the USA) for the beginning of March 2003 to the beginning of March 2008.



These figures might lead us to expect that by the beginning of March 2009 the index would be perhaps as high as 12500. In fact it had fallen to just a little over 7000.

Events causing results that seem out of line with a suggested trend may not always be unexpected. The graph on the next page shows the total expenditure, in billions of pounds, of the people visiting London, from elsewhere in the UK, for the years 2008 to 2012.



Source of data: London tourism report 2012/13. London & Partners.

Does the somewhat higher than trend figure for 2012 indicate increasingly higher figures for 2013, 2014 etc. are likely, or should it be expected with 2012 being the year of the London Olympics?

Exercise 1A

1. AGEING POPULATION.
The table below shows some real and some predicted figures for the number of persons over 65, per 100 persons of working age, in Australia for 1971, 1981, 1991, 2001, 2011, 2021 and 2031.

Number of persons over 65 per 100 persons of working age.							
Year	1971	1981	1991	2001	2011	2021	2031
Number	13	15	17	18	20	26	31

Source of data: HBF.

(a) We would of course expect the number of people over 65 to increase as the population of Australia increases. Can the increase in the numbers shown in the bottom row of the table (i.e. 13, 15, 17, 18, ...) be attributed to this natural increase in the population or not? Explain your answer.

(b) Display the data as a line graph.

(c) Describe any trends shown.

2. POPULATION OF WESTERN AUSTRALIA.

The following table gives the population (to the nearest 1000) of Western Australia in various years.

Year	1915	1925	1935	1945	1955
Population	317 000	378 000	450 000	490 000	669 000

Year	1965	1975	1985	1995	2005
Population	838 000	1 167 000	1 437 000	1 749 000	2 037 000

[Based on Australian Bureau of Statistics data.]

(a) Display this information as a line graph.

(b) Comment on any trends shown.

(c) By "eyeballing" your graph, predict what the population was in the year 1950 and comment on the likely reliability of your prediction.

(d) By "eyeballing" your graph, predict what the population will be in the year 2025 and comment on the likely reliability of your prediction.

(e) Find out the latest population figure for Western Australia and see how it compares to that predicted by the above figures.

3. VISITORS TO AUSTRALIA.

The table below shows the number of overseas visitors arriving in Australia for a stay of less than 12 months (called *short term visitors*) for various years from 1992 to 2013.

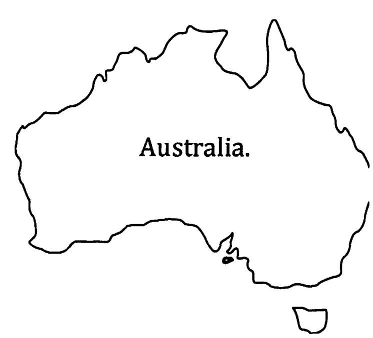
Year	1992	1994	1997	2001	2003	2005
Visitors	2 603 100	3 361 600	4 318 000	4 855 800	4 745 800	5 463 000

Year	2007	2009	2010	2011	2012	2013
Visitors	5 588 800	5 490 200	5 790 100	5 770 800	6 032 200	6 380 500

[Based on Australian Bureau of Statistics data.]

(a) Display this information as a line graph and comment on any trends shown.

(b) By "eyeballing" your graph, predict the number of short term visitors to Australia in 1999 and in 2020, in each case commenting on the likely reliability of your prediction.



4. AGE AT FIRST MARRIAGE.

In more recent times is the age at which people get married for the first time more or less than it used to be in "the old days"?

Use the data shown below to formulate a response to the above question, including in your response a graphical display of the data, and commenting on, and describing, any trends shown.

Median age at first marriage in Australia.

Year	1966	1970	1974	1980	1982	1984	1985	1988	1990	1991	1992
Men	23·8	23·4	23·3	24·2	24·6	25·1	25·4	26·1	26·5	26·7	26·9
Women	21·2	21·1	20·9	21·9	22·4	22·9	23·2	24·0	24·3	24·5	24·7

Year	1993	1996	1997	2000	2001	2002	2008	2009	2010	2011	2012
Men	27·0	27·6	27·8	28·5	28·7	29·0	29·6	29·6	29·6	29·7	29·8
Women	24·8	25·7	25·9	26·7	26·9	27·1	27·7	27·7	27·9	28·0	28·1

[Source of data: Australian Bureau of Statistics.]

5. NUMBER OF MARRIAGES.

Let us suppose that in a particular country, the number of marriages occurring in various years from 1910 to 2010 were as shown in the following table.

Number of Marriages occurring in the country in various years from 1910 – 2010

Year (Y)	1910	1925	1930	1938	1947	1962	1975	1986	1997	2004	2010
Number (N)	25124	34728	37894	38247	46221	49241	59258	66241	68256	71567	76278

(a) View the graph of this time series data on a calculator or computer and confirm that the points are suitable for linear modelling.

(b) With N representing the number of marriages in a year, and Y representing the year, obtain the least squares regression line, $N = mY + b$, for this data, giving m to the nearest integer and b to the nearest 1000.

6. EXPANDING BUSINESS.

The rental property manager of a real estate business researched data regarding the number of properties managed by the business at various times from 2010 to 2015. The table below shows the data she collected.

Month & Year	Mar 2010	Mar 2011	Mar 2012	Mar 2013	Mar 2014	Jun 2014	Sept 2014	Dec 2014	Mar 2015	Jun 2015
N$^{o.}$ of Properties Managed	302	321	408	467	490	492	508	502	526	540

(a) Commencing with March 2010 as $t = 1$, June 2010 as $t = 2$ etc determine the equation of the least squares linear regression line for predicting the number of properties managed for a given value of t.

(b) Assuming this trend continues predict the number of properties managed by the group in December 2017.

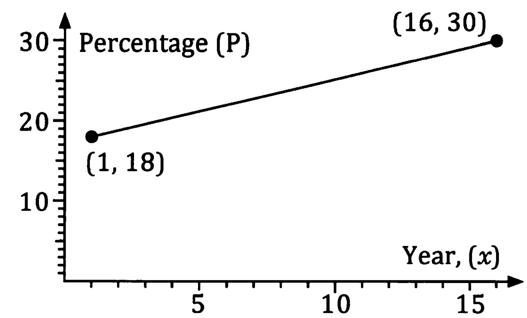
7. CAESAREAN SECTIONS.
Let us suppose that in a particular region the number of deliveries for all births, over a sixteen year period, and the number of those deliveries that were performed by caesarean section, were as given in the following table:

Year	1	2	3	4	5	6	7	8
Total births	15 750	16 170	15 971	16 227	16 451	16 332	16 815	16 213
Caesarean	2 835	3 017	2 925	2 973	3 068	3 612	3 821	3 659

Year	9	10	11	12	13	14	15	16
Total births	16 629	16 873	17 105	16 974	17 043	16 142	17 123	17 260
Caesarean	4 083	4 431	4 862	4 904	5082	4 758	5 161	5 178

In year 1 the rate of caesarean deliveries was 2 835 out of 15 750, ie. 18%.
In year 16 the rate of caesarean deliveries was 5 178 out of 17 260, ie. 30%.

If this increase in the caesarean rate over this time were perfectly linear the caesarean rates would fit the graph shown on the right.



(a) Determine the equation of this straight line, in the form P = mx + c, with x and P as indicated on the graph, and m and c constants.

(b) Plot a graph of the real percentage figures over the 16 years to show that the points do not form such a straight line and describe any trends the real percentage figures show over time.

8. Transport authorities in a particular country researched vehicle registrations over a number of years to investigate the average age of the vehicles registered for road use. The results were:

AVERAGE AGE OF VEHICLES							
YEAR	1990	1992	1994	1996	1998	2000	2002
AVERAGE AGE (Years)	7·6	7·8	8·4	9·1	9·6	10·2	10·5
YEAR	2004	2006	2008	2010	2012	2014	
AVERAGE AGE (Years)	10·6	10·7	10·7	10·6	10·4	9·9	

(a) Display this data as a time series graph.
(b) Is the data suitable for linear regression? Justify your answer.
(c) Use your graph to predict the average age of vehicles in this country for the years 1999 and 2020, commenting in each case about the likely reliability of your prediction.

9. Confirm that the following paired values are not suited to linear regression.

t	1	2	3	4	5	6	7	8	9	10
Y	5	20	44	96	149	217	310	410	525	661

Explore the ability of graphic calculators to determine non linear regression models to obtain a model for the above data of the form $Y = a \times t^b$.

THE CHANGING SEASONS

Quite a number of middle aged or elderly people have been heard to comment that they feel the seasons have been changing over the years. For example, some feel that January tended to be hotter in the past. Are they correct or is it that they are only remembering the more memorable summers?

The table below shows information about January temperatures recorded at a particular location in Perth over a period of approximately 70 years. Does the information given in the table suggest that, during the period of time the statistics cover, summer temperatures in Perth were showing some changing pattern?

Write a brief report justifying your opinion and include suitable graphs based on the information in the table below.

TEMPERATURE STATISTICS FOR THE MONTH OF JANUARY 1946 - 2014

Year	1946	1948	1950	1952	1954
Mean daily max temp (°C)	28·2	30·9	32·7	31·3	31·0
Mean daily min temp (°C)	13·3	15·3	16·8	14·8	17·6

Year	1956	1958	1960	1962	1964
Mean daily max temp (°C)	33·8	33·1	28·2	35·6	30·6
Mean daily min temp (°C)	17·1	18·2	14·8	19·5	16·7

Year	1966	1968	1970	1972	1974
Mean daily max temp (°C)	31·1	30·5	29·3	32·7	34·2
Mean daily min temp (°C)	17·4	17·5	16·7	16·5	17·4

Year	1976	1978	1980	1982	1984
Mean daily max temp (°C)	30·8	34·0	34·0	29·4	32·4
Mean daily min temp (°C)	16·9	18·4	18·3	15·9	16·8

Year	1986	1988	1990	1992	1994
Mean daily max temp (°C)	33·5	30·2	29·1	33·5	30·9
Mean daily min temp (°C)	18·6	15·5	16·1	19·2	17·1

Year	1996	1998	2000	2002	2004
Mean daily max temp (°C)	31·5	32·6	30·7	31·8	32·4
Mean daily min temp (°C)	16·8	17·3	17·8	16·4	17·6

Year	2006	2008	2010	2012	2014
Mean daily max temp (°C)	29·9	33·8	35·0	33·4	32·9
Mean daily min temp (°C)	16·8	18·8	18·8	19·7	17·8

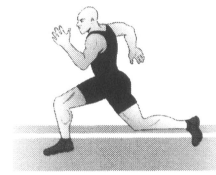
[Source: Bureau of Meteorology.]

Research and analysis.

Research and obtain appropriate time series data to answer one of the following questions, or negotiate with you teacher an alternative question of your choice that will similarly require an analysis of time series data.

Your answer should include the data you obtained, suitably presented, an explanation of the analysis of that data and a reasoned conclusion that addresses the question.

When is the world record for running 100 metres likely to fall to 9 seconds?



When is the population of the world likely to reach 10 billion?

What is the population of China likely to be in the year 2050?

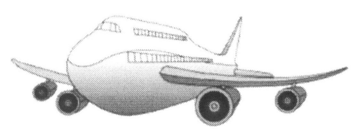
What trends does Australia's unemployment rate show over time?

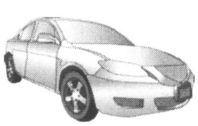
How has the share price of a particular company of your choice varied with time and what might this suggest for the future price?

How have house prices, or rental costs, varied over time in your area and what does this suggest for future price, or rental, movements?

How has the price of farming commodities varied with time and what does this suggest for future price movements?

How has the volume of air traffic changed with time and what does this suggest for future air traffic volume?





How have national or worldwide annual sales of new cars changed with time and what does this suggest for future numbers of sales?

How have the estimated numbers of some endangered species of animal varied with time and what does this suggest for the future of the species?

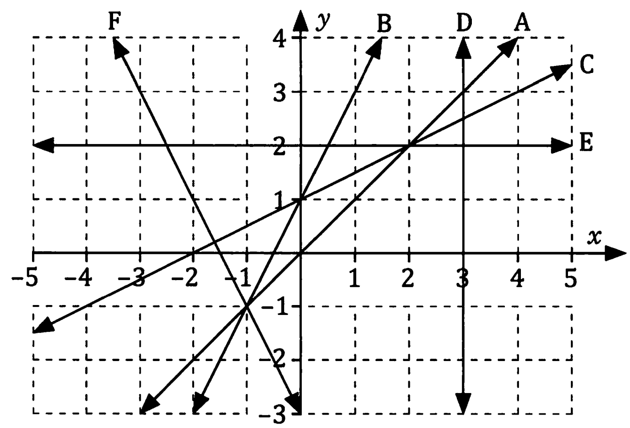
Miscellaneous Exercise One.

This miscellaneous exercise may include questions involving the work of this chapter and the ideas mentioned in the preliminary work section at the beginning of the book.

1. Increase \$2050 by 13%.

2. Decrease \$12600 by 74%

3. Determine the equation of each of the straight lines A to F shown in the graph on the right.



4. Find the first five terms of each of the following recursively defined sequences.
   (a) $T_1 = 25$, $T_{n+1} = T_n + 2$.
   (b) $T_1 = 32$, $T_{n+1} = T_n - 1{\cdot}5$
   (c) $T_1 = 64$, $T_{n+1} = 1{\cdot}5 \times T_n$.
   (d) $T_1 = 4000$, $T_n = T_{n-1} \div 2$
   (e) $T_1 = 5$, $T_n = 2T_{n-1} + 7$.
   (f) $T_1 = 16$, $T_{n+1} = 1{\cdot}5T_n - 4$

5. POPULATION CHANGES.
   The table on the right shows the population of Perth, and of Western Australia as a whole, in each of the census years from 1911 to 1991.
   Make a table of your own showing Perth's population as a percentage of the WA population for each of these census years.
   (a) Plot these percentages as a line graph, with years on the horizontal axis, and comment on the trends your graph shows.
   (b) Use your graph to suggest Perth's population as a percentage of the WA population for (i) 1940 and (ii) 2020 in each case commenting on the likely reliability of your estimates.

Population of WA and Perth.
Source: Australian Bureau of Statistics

Year	W.A.	Perth
1911	282114	116181
1921	332732	170213
1933	438852	230340
1947	502480	302968
1954	639771	395049
1961	736629	475398
1966	836673	558821
1971	1030469	703199
1976	1178340	832760
1981	1300600	922017
1986	1459011	1050120
1991	1636067	1188762

Try to find more recent data for the population of Perth and of Western Australia, say for the census year 2011, and comment on how it compares with the data given here.